

PSYCH254 HW3

2026-01-30

1) Estimate an unconditional (empty) model for the three level-1 variables in the model. This can be done in a single analysis (see lecture slides). Because frlunch is binary, you will need to list this variable after the nominal command to invoke an appropriate categorical variable (logistic) model. The nominal commands are as follows.

```
* nominal: frlunch; # Blimp Studio
```

```
* nominal = 'frlunch', # rblimp
```

Obtain the intraclass correlation (ICC) for each level-1 predictor. Blimp prints these in the effect size section of each summary table. Type the values of the ICCs into the quiz question's text field. (1 point)

```
p1_mod <- rblimp(  
  data = df,  
  clusterid = 'school',  
  nominal = 'frlunch',  
  model = '  
    psolvepre ~ intercept | intercept;  
    frlunch ~ intercept | intercept',  
  seed = 90291,  
  burn = 10000,  
  iter = 10000  
)
```

```
## > Checking for Blimp Update...
```

```
output(p1_mod)
```

```
##  
## -----  
##  
##                               Blimp  
##                               4.0.2  
##  
##           Blimp was developed with funding from Institute of  
##           Education Sciences awards R305D150056 and R305D190002.  
##  
##           Craig K. Enders, P.I. Email: cenders@psych.ucla.edu  
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##           Han Du, Co-P.I. Email: hdu@psych.ucla.edu  
##           Roy Levy, Co-P.I. Email: roy.levy@asu.edu
```

```

##
##           Programming and Blimp Studio by Brian T. Keller
##
##           There is no expressed license given.
## -----
##
##
## ALGORITHMIC OPTIONS SPECIFIED:
##
##   Imputation method:           Fully Bayesian model-based
##   MCMC algorithm:              Full conditional Metropolis sampler with
##                               Auto-Derived Conditional Distributions
##   Default Priors:
##     Multilevel covariance matrices: Zero matrix, df = -(p + 1) (PRIOR2)
##     Outcome covariance matrices:   Zero matrix, df = -(p + 1) (PRIOR2)
##     Predictor model variances:     Unit sum of squares, df = 2 (XPRIOR1)
##   Chain Starting Values:         Random starting values
##
##
## NOTE: The default prior for regression coefficients
##       in categorical models is 'normal( 0.0, 5.0)'
##
## NOTE: For complete list of priors include 'pinf' in the options.
##
## NOTE: The following nominal variables have automatically
##       been assigned a reference group.
##
##       'frlunch' with reference value '0'
##
##       To change reference group use the following command:
##       NOMINAL: variable_name(reference_value);
##
##
## BURN-IN POTENTIAL SCALE REDUCTION (PSR) OUTPUT:
##
## NOTE: Split chain PSR is being used. This splits each chain's
##       iterations to create twice as many chains.
##
## Comparing iterations across 2 chains   Highest PSR   Parameter #
##           251 to 500                   1.055         10
##           501 to 1000                   1.091          4
##           751 to 1500                   1.025          2
##          1001 to 2000                   1.018         10
##          1251 to 2500                   1.019          4
##          1501 to 3000                   1.015         10
##          1751 to 3500                   1.015          4
##          2001 to 4000                   1.003          1
##          2251 to 4500                   1.009         10
##          2501 to 5000                   1.005         10
##          2751 to 5500                   1.004          2
##          3001 to 6000                   1.013          2
##          3251 to 6500                   1.008          2
##          3501 to 7000                   1.002          2

```

```

##          3751 to 7500          1.002          2
##          4001 to 8000          1.006          2
##          4251 to 8500          1.005          2
##          4501 to 9000          1.007          4
##          4751 to 9500          1.009          2
##          5001 to 10000         1.004          2
##
##
## DATA INFORMATION:
##
##   Level-2 identifier:      school
##   Sample Size:            982
##   Level-2 Clusters:       29
##   Nominal Dummy Codes:
##
##           frlunch = frlunch.1
##   Missing Data Info:
##
##           Pattern #:          1
##           miss % -----
##           frlunch =  0.0      -
##           psolvepre = 0.0     -
##           -----
##           % 100.0
##
##
## MODEL INFORMATION:
##
##   NUMBER OF PARAMETERS
##     Outcome Models:        5
##     Predictor Models:       0
##
##   MODELS
##     [1] frlunch ~ Intercept
##     [2] psolvepre ~ Intercept
##
##
## WARNING MESSAGES:
##
##   No warning messages.
##
##
## MODEL FIT:
##
##
##   INFORMATION CRITERIA
##
##     Conditional Likelihood
##       DIC2                6699.706
##       WAIC                 6718.525
##
##
## IMPUTED DATA SUMMARIES:
##

```

NOTE: Imputed data summaries are averaged across 10000 data sets.

##

##

## Variables	Mean	StdDev	25%	50%	75%	Min
##	-----					
## frlunch.1	0.801	0.399	1.000	1.000	1.000	0.000
## psolvepre	49.910	4.945	47.000	50.000	53.000	37.000

##

##

##

##

##

CORRELATIONS AMONG RESIDUALS:

##

Summaries based on 10000 iterations using 2 chains.

NOTE: Estimate column based on posterior median.

##

##

Level-1:

##

## Correlations	Estimate	StdDev	2.5%	97.5%	ChiSq	PValue
##	-----					
## frlunch.1, psolvepre	-0.000	0.037	-0.073	0.074	0.000	0.999

##

##

##

##

##

OUTCOME MODEL ESTIMATES:

##

Summaries based on 10000 iterations using 2 chains.

NOTE: Estimate column based on posterior median.

##

##

Outcome Variable: frlunch

##

## Parameters	Estimate	StdDev	2.5%	97.5%	ChiSq	PValue
##	-----					
## frlunch.1						

##

Variances:

## L2 : Var(Intercept)	0.840	0.369	0.411	1.813	---	---
------------------------	-------	-------	-------	-------	-----	-----

##

Coefficients:

## Intercept	1.646	0.209	1.247	2.073	62.561	0.000
--------------	-------	-------	-------	-------	--------	-------

##

Standard Deviations:

## L2 : SD(Intercept)	0.916	0.182	0.641	1.346	---	---
-----------------------	-------	-------	-------	-------	-----	-----

##

Odds Ratio:

## Intercept	5.187	1.132	3.479	7.950	---	---
--------------	-------	-------	-------	-------	-----	-----

##

```

## Proportion Variance Explained
##   by Coefficients           0.000      0.000      0.000      0.000      ---      ---
##   by Level-2 Random Intercepts 0.203      0.064      0.111      0.355      ---      ---
##   by Level-1 Residual Variation 0.797      0.064      0.645      0.889      ---      ---
##
## -----
##
##
## Outcome Variable:  psolvepre
##
## Parameters           Estimate      StdDev      2.5%      97.5%      ChiSq      PValue
## -----
## Variances:
##   L2 : Var(Intercept)      2.831      1.126      1.473      5.753      ---      ---
##   Residual Var.           21.893      1.011     20.009     23.980      ---      ---
##
## Coefficients:
##   Intercept              49.857      0.367     49.152     50.599    18438.449      0.000
##
## Standard Deviations:
##   L2 : SD(Intercept)       1.682      0.307      1.214      2.399      ---      ---
##   Residual SD              4.679      0.108      4.473      4.897      ---      ---
##
## Proportion Variance Explained
##   by Coefficients           0.000      0.000      0.000      0.000      ---      ---
##   by Level-2 Random Intercepts 0.114      0.038      0.063      0.209      ---      ---
##   by Level-1 Residual Variation 0.886      0.038      0.791      0.937      ---      ---
##
## -----

```

The ICC for each level-1 predictor is: 0.797 and 0.886.

2) Consider the ICC for the dummy coded lunch assistance variable, *frlunch*. The ICC for this variable is relatively large. When the level-1 predictor is binary as it is here, what is the between-cluster variation of the variable measuring? (2 points)

When the level-1 predictor is binary, the between-cluster variation of the variable is measuring the differences in the log-odds of the event across schools. Essentially, it represents the between-school differences in the proportion of student receiving free or reduced price lunch.

3) Fit a model that includes the two level-1 predictors. Center the predictors at their school means (i.e., centering within cluster, group mean centering). Using the simplified notation from class, the model is shown below.

$$psolvepost_{ij} = \beta_0 + \beta_1(psolvepre_{ij}^w) + \beta_2(frlunch_{ij}^w) + u_{0j} + \epsilon_{ij}$$

In Blimp, you can automatically create a dummy code by listing frlunch after the nominal command. Blimp's group mean centering function works the same for categorical and continuous predictors. Upload a screen shot of your regression summary table into the BruinLearn quiz. (1 point)

```
p3_mod <- rblimp(
  data = df,
  cluster = 'school',
  center = 'groupmean = psolvepre frlunch',
  nominal = 'frlunch',
  model = '
psolvepost ~ intercept psolvepre frlunch | intercept',
  seed = 90291,
  burn = 10000,
  iter = 10000
)

output(p3_mod)
```

```
##
## -----
##
##                               Blimp
##                               4.0.2
##
##       Blimp was developed with funding from Institute of
##       Education Sciences awards R305D150056 and R305D190002.
##
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##       Han Du, Co-P.I. Email: hdu@psych.ucla.edu
##       Roy Levy, Co-P.I. Email: roy.levy@asu.edu
##
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##
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##
## -----
##
##
## ALGORITHMIC OPTIONS SPECIFIED:
##
##   Imputation method:           Fully Bayesian model-based
##   MCMC algorithm:              Full conditional Metropolis sampler with
##                               Auto-Derived Conditional Distributions
##
##   Default Priors:
##     Multilevel covariance matrices: Zero matrix, df = -(p + 1) (PRIOR2)
##     Outcome covariance matrices:   Zero matrix, df = -(p + 1) (PRIOR2)
##     Predictor model variances:     Unit sum of squares, df = 2 (XPRIOR1)
##   Chain Starting Values:        Random starting values
##
##
## NOTE: The default prior for regression coefficients
```

```

##          in categorical models is 'normal( 0.0, 5.0)'
##
## NOTE: For complete list of priors include 'pinf' in the options.
##
## NOTE: The following nominal variables have automatically
##        been assigned a reference group.
##
##        'frlunch' with reference value '0'
##
##        To change reference group use the following command:
##        NOMINAL: variable_name(reference_value);
##
## BURN-IN POTENTIAL SCALE REDUCTION (PSR) OUTPUT:
##
## NOTE: Split chain PSR is being used. This splits each chain's
##        iterations to create twice as many chains.
##
## Comparing iterations across 2 chains      Highest PSR      Parameter #
##              251 to 500                    1.053              3
##              501 to 1000                   1.046              3
##              751 to 1500                   1.030              3
##             1001 to 2000                   1.105              3
##             1251 to 2500                   1.026             22
##             1501 to 3000                   1.042              3
##             1751 to 3500                   1.042              3
##             2001 to 4000                   1.012              3
##             2251 to 4500                   1.016              3
##             2501 to 5000                   1.011             17
##             2751 to 5500                   1.012              3
##             3001 to 6000                   1.006              3
##             3251 to 6500                   1.004             17
##             3501 to 7000                   1.004             17
##             3751 to 7500                   1.006              3
##             4001 to 8000                   1.005              3
##             4251 to 8500                   1.009              3
##             4501 to 9000                   1.004              3
##             4751 to 9500                   1.011              3
##             5001 to 10000                  1.004              3
##
##
## METROPOLIS-HASTINGS ACCEPTANCE RATES:
##
## Chain 1:
##
## Variable                Type      Probability      Target Value
## frlunch                  latent imp      0.488          0.500
## psolvepre                 latent imp      0.505          0.500
##
## NOTE: Suppressing printing of 1 chains.
##        Use keyword 'tuneinfo' in options to override.
##
## DATA INFORMATION:

```

```

##
## Level-2 identifier:      school
## Sample Size:           982
## Level-2 Clusters:      29
## Nominal Dummy Codes:
##
##           frlunch = frlunch.1
## Missing Data Info:
##
##           Pattern #:      1
##           miss % -----
##           psolvepost = 0.0      -
##           frlunch = 0.0      -
##           psolvepre = 0.0      -
##           -----
##           % 100.0
##
##
## MODEL INFORMATION:
##
## NUMBER OF PARAMETERS
## Outcome Models:      5
## Predictor Models:    9
##
## PREDICTORS
## Complete continuous:  psolvepre
## Complete nominal:    frlunch
##
## CENTERED PREDICTORS
## Group Mean Centered:  frlunch psolvepre
##
## MODELS
## [1] psolvepost ~ Intercept psolvepre frlunch.1
##
## WARNING MESSAGES:
##
## No warning messages.
##
## MODEL FIT:
##
## INFORMATION CRITERIA
##
## Marginal Likelihood
## DIC2      5815.830
## WAIC      5816.473
##
## Conditional Likelihood
## DIC2      5749.134
## WAIC      5760.469
##
##

```

``` ## IMPUTED DATA SUMMARIES: ```

```
##
```

```
## NOTE: Imputed data summaries are averaged across 10000 data sets.
```

```
##
```

```
##
```

## Variables	Mean	StdDev	25%	50%	75%	Min
##	-----					
## psolvepost	54.009	5.749	51.000	54.000	58.000	37.000
## frlunch.1	0.801	0.399	1.000	1.000	1.000	0.000
## psolvepre	49.910	4.945	47.000	50.000	53.000	37.000

```
##
```

```
##
```

```
##
```

```
##
```

```
##
```

``` ## CORRELATIONS AMONG RESIDUALS: ```

```
##
```

```
## No residual correlations.
```

```
##
```

```
##
```

```
##
```

``` ## OUTCOME MODEL ESTIMATES: ```

```
##
```

```
## Summaries based on 10000 iterations using 2 chains.
```

```
## NOTE: Estimate column based on posterior median.
```

```
##
```

```
##
```

```
## Outcome Variable: psolvepost
```

```
##
```

```
## Group Mean Centered: frlunch.1 psolvepre
```

```
##
```

```
##
```

## Parameters	Estimate	StdDev	2.5%	97.5%	ChiSq	PValue
##	-----					

```
##
```

``` ## Variances: ```

## L2 : Var(Intercept)	8.085	2.805	4.560	15.210	---	---
## Residual Var.	20.199	0.918	18.498	22.089	---	---

```
##
```

``` ## Coefficients: ```

## Intercept	53.754	0.557	52.697	54.880	9307.682	0.000
## psolvepre	0.463	0.031	0.402	0.524	217.188	0.000
## frlunch.1	-0.574	0.394	-1.361	0.196	2.162	0.141

```
##
```

``` ## Standard Deviations: ```

## L2 : SD(Intercept)	2.843	0.457	2.135	3.900	---	---
## Residual SD	4.494	0.102	4.301	4.700	---	---

```
##
```

``` ## Standardized Coefficients: ```

## psolvepre	0.375	0.027	0.320	0.426	190.579	0.000
## frlunch.1	-0.037	0.025	-0.088	0.013	2.157	0.142

```
##
```

``` ## Proportion Variance Explained ```

## by Coefficients	0.142	0.020	0.104	0.184	---	---
--------------------	-------	-------	-------	-------	-----	-----

```

## by Level-2 Random Intercepts      0.245      0.059      0.153      0.380      ---      ---
## by Level-1 Residual Variation      0.610      0.050      0.500      0.693      ---      ---
##
## -----
##
##
##
## PREDICTOR MODEL ESTIMATES:
##
## Summaries based on 10000 iterations using 2 chains.
## NOTE: Estimate column based on posterior median.
##
##
## Complete predictor: frlunch
##
## Parameters      Estimate      StdDev      2.5%      97.5%      ChiSq      PValue
## -----
## Grand Mean: frlunch.1      0.976      0.123      0.746      1.228      63.753      0.000
##
## Level 1:
##   frlunch.1 ~ psolvepre      0.001      0.010      -0.020      0.021      0.005      0.946
##   Var(frlunch.1)      1.000      0.000      1.000      1.000      ---      ---
##
## Level 2:
##   frlunch.1 ~ psolvepre      -0.177      0.080      -0.343      -0.024      4.969      0.026
##   Var(frlunch.1)      0.224      0.086      0.117      0.447      ---      ---
##
## -----
##
##
## Complete predictor: psolvepre
##
## Parameters      Estimate      StdDev      2.5%      97.5%      ChiSq      PValue
## -----
## Grand Mean      49.846      0.352      49.142      50.538      20003.776      0.000
##
## Level 1:
##   frlunch.1      0.016      0.230      -0.438      0.468      0.005      0.945
##   Residual Var.      21.828      1.000      19.990      23.952      ---      ---
##
## Level 2:
##   frlunch.1      -1.523      0.642      -2.831      -0.248      5.642      0.018
##   Residual Var.      1.666      0.697      0.784      3.482      ---      ---
##
## -----

```

4) Fit a model that includes the two level-1 predictors (within-cluster or group mean centered), their level-2 school means, and the school-level covariate, *eslpct*. In Blimp, you can add the level-2 means using the *.mean* function, as shown in the lecture slides. Center all level-2 variables at their grand means to define the intercept as the post-test grand mean. Using simplified notation from class, the model is shown below. Upload a screen shot of your regression summary table into the BruinLearn quiz. (1 point)

$$psolvepost_{ij} = \beta_0 + \beta_1(psolvepre_{ij}^w) + \beta_2(frlunch_{ij}^w) + \beta_3(psolvepre_j^b) + \beta_4(frlunch_j^b) + \beta_5(eslpct_j) + u_{0j} + \epsilon_{ij}$$

```
p4_mod <- rblimp(
  data = df,
  cluster = 'school',
  center = '
  groupmean = psolvepre frlunch;
  grandmean = psolvepre.mean frlunch.mean',
  nominal = 'frlunch',
  model = '
  psolvepost ~ intercept psolvepre psolvepre.mean frlunch frlunch.mean eslpct | intercept',
  seed = 90291,
  burn = 10000,
  iter = 10000
)

output(p4_mod)
```

```
##
## -----
##
##                               Blimp
##                               4.0.2
##
##       Blimp was developed with funding from Institute of
##       Education Sciences awards R305D150056 and R305D190002.
##
##       Craig K. Enders, P.I. Email: cenders@psych.ucla.edu
##       Brian T. Keller, Co-P.I. Email: btkeller@missouri.edu
##       Han Du, Co-P.I. Email: hdu@psych.ucla.edu
##       Roy Levy, Co-P.I. Email: roy.levy@asu.edu
##
##       Programming and Blimp Studio by Brian T. Keller
##
##       There is no expressed license given.
##
## -----
##
##
## ALGORITHMIC OPTIONS SPECIFIED:
##
##   Imputation method:           Fully Bayesian model-based
##   MCMC algorithm:              Full conditional Metropolis sampler with
##                               Auto-Derived Conditional Distributions
##
##   Default Priors:
##   Multilevel covariance matrices: Zero matrix, df = -(p + 1) (PRIOR2)
```

```

## Outcome covariance matrices: Zero matrix, df = -(p + 1) (PRIOR2)
## Predictor model variances: Unit sum of squares, df = 2 (XPRIOR1)
## Chain Starting Values: Random starting values
##
##
## NOTE: The default prior for regression coefficients
## in categorical models is 'normal( 0.0, 5.0)'
##
## NOTE: For complete list of priors include 'pinf' in the options.
##
## NOTE: The following nominal variables have automatically
## been assigned a reference group.
##
## 'frlunch' with reference value '0'
##
## To change reference group use the following command:
## NOMINAL: variable_name(reference_value);
##
## BURN-IN POTENTIAL SCALE REDUCTION (PSR) OUTPUT:
##
## NOTE: Split chain PSR is being used. This splits each chain's
## iterations to create twice as many chains.
##
## Comparing iterations across 2 chains Highest PSR Parameter #
## 251 to 500 1.198 29
## 501 to 1000 1.131 29
## 751 to 1500 1.206 29
## 1001 to 2000 1.086 29
## 1251 to 2500 1.056 29
## 1501 to 3000 1.067 29
## 1751 to 3500 1.053 29
## 2001 to 4000 1.047 29
## 2251 to 4500 1.032 29
## 2501 to 5000 1.020 29
## 2751 to 5500 1.017 29
## 3001 to 6000 1.009 3
## 3251 to 6500 1.019 29
## 3501 to 7000 1.010 29
## 3751 to 7500 1.016 3
## 4001 to 8000 1.009 3
## 4251 to 8500 1.010 23
## 4501 to 9000 1.014 29
## 4751 to 9500 1.020 29
## 5001 to 10000 1.015 29
##
## METROPOLIS-HASTINGS ACCEPTANCE RATES:
##
## Chain 1:
##
## Variable Type Probability Target Value
## frlunch latent imp 0.502 0.500
## frlunch parameter 0.469 0.500

```

```

##      psolvepre          latent imp          0.493          0.500
##      psolvepre          parameter          0.499          0.500
##
## NOTE: Suppressing printing of 1 chains.
##       Use keyword 'tuneinfo' in options to override.
##
##
## DATA INFORMATION:
##
## Level-2 identifier:      school
## Sample Size:            982
## Level-2 Clusters:       29
## Nominal Dummy Codes:
##
##           frlunch = frlunch.1
## Missing Data Info:
##
##           Pattern #:      1
##           miss % -----
##           psolvepost = 0.0   -
##           eslpct = 0.0      -
##           frlunch = 0.0      -
##           psolvepre = 0.0      -
##           -----
##           % 100.0
##
##
## MODEL INFORMATION:
##
## NUMBER OF PARAMETERS
## Outcome Models:         8
## Predictor Models:       15
##
## PREDICTORS
## Complete continuous:     eslpct psolvepre
## Complete nominal:       frlunch
##
## CENTERED PREDICTORS
## Group Mean Centered:    frlunch psolvepre
##
## LATENT CLUSTER MEANS
## Grand Mean Centered:    frlunch psolvepre
##
## MODELS
## [1] psolvepost ~ Intercept psolvepre psolvepre.mean[school] frlunch.1
##           frlunch.1.mean[school] eslpct
##
##
## WARNING MESSAGES:
##
## No warning messages.
##
##
## MODEL FIT:

```

```

##
##
## INFORMATION CRITERIA
##
## Marginal Likelihood
## DIC2 5804.606
## WAIC 5806.502
##
## Conditional Likelihood
## DIC2 5751.040
## WAIC 5764.029
##
##
## IMPUTED DATA SUMMARIES:
##
## NOTE: Imputed data summaries are averaged across 10000 data sets.
##
##
## Variables Mean StdDev 25% 50% 75% Min
## -----
## psolvepost 54.009 5.749 51.000 54.000 58.000 37.000
## eslpct 54.172 22.739 39.000 58.000 67.750 10.000
## frlunch.1 0.801 0.399 1.000 1.000 1.000 0.000
## psolvepre 49.910 4.945 47.000 50.000 53.000 37.000
## -----
##
##
## CORRELATIONS AMONG RESIDUALS:
##
## No residual correlations.
##
##
## OUTCOME MODEL ESTIMATES:
##
## Summaries based on 10000 iterations using 2 chains.
## NOTE: Estimate column based on posterior median.
##
##
## Outcome Variable: psolvepost
##
## Grand Mean Centered: frlunch.1.mean[school] psolvepre.mean[school]
## Group Mean Centered: frlunch.1 psolvepre
##
##
## Parameters Estimate StdDev 2.5% 97.5% ChiSq PValue
## -----
## Variances:
## L2 : Var(Intercept) 4.765 1.945 2.415 9.830 --- ---
## Residual Var. 20.196 0.941 18.467 22.143 --- ---
##

```

```

## Coefficients:
##   Intercept          53.300      1.425    50.607    56.166   1400.449    0.000
##   psolvepre           0.462      0.031     0.400     0.521    219.573    0.000
##   psolvepre.mean[school] 1.044      0.404     0.261     1.879     6.713    0.010
##   frlunch.1          -0.598      0.392    -1.381     0.154     2.361    0.124
##   frlunch.1.mean[school] -3.880      3.829   -10.976     4.169     0.962    0.327
##   eslpct              0.009      0.024    -0.041     0.053     0.114    0.735
##
## Standard Deviations:
##   L2 : SD(Intercept)      2.183      0.406     1.554     3.135      ---      ---
##   Residual SD            4.494      0.104     4.297     4.706      ---      ---
##
## Standardized Coefficients:
##   psolvepre              0.367      0.026     0.315     0.416    195.397    0.000
##   psolvepre.mean[school] 0.286      0.102     0.073     0.471     7.657    0.006
##   frlunch.1             -0.038      0.025    -0.087     0.010     2.359    0.125
##   frlunch.1.mean[school] -0.096      0.093    -0.266     0.101     0.980    0.322
##   eslpct                 0.031      0.083    -0.141     0.186     0.119    0.730
##
## Proportion Variance Explained
##   by Coefficients          0.267      0.045     0.184     0.356      ---      ---
##   by Level-2 Random Intercepts 0.140      0.046     0.074     0.252      ---      ---
##   by Level-1 Residual Variation 0.589      0.047     0.486     0.669      ---      ---
##
## -----
##
##
##
##
## PREDICTOR MODEL ESTIMATES:
##
##   Summaries based on 10000 iterations using 2 chains.
##   NOTE: Estimate column based on posterior median.
##
##
## Complete predictor: eslpct
##
## Parameters          Estimate      StdDev      2.5%      97.5%      ChiSq      PValue
## -----
## Grand Mean          54.162      4.588      45.167      63.250    139.341    0.000
##
## Level 2:
##   frlunch.1          12.321      12.934     -13.125      37.842     0.911     0.340
##   psolvepre           6.121       4.116      -1.896      14.495     2.235     0.135
##   Residual Var.      439.597     146.983    245.845     813.164      ---      ---
##
## -----
##
##
##
## Complete predictor: frlunch
##

```

## Parameters	Estimate	StdDev	2.5%	97.5%	ChiSq	PValue
##						
##						
## Grand Mean: frlunch.1	0.985	0.127	0.734	1.238	60.609	0.000
##						
## Level 1:						
## frlunch.1 ~ psolvepre	0.001	0.011	-0.019	0.022	0.007	0.934
## Var(frlunch.1)	1.000	0.000	1.000	1.000	---	---
##						
## Level 2:						
## frlunch.1 ~ eslpct	0.005	0.005	-0.006	0.016	0.837	0.360
## frlunch.1 ~ psolvepre	-0.195	0.082	-0.363	-0.037	5.689	0.017
## Var(frlunch.1)	0.214	0.091	0.105	0.448	---	---
##						
##						
##						
##						
## Complete predictor: psolvepre						
##						
## Parameters	Estimate	StdDev	2.5%	97.5%	ChiSq	PValue
##						
##						
## Grand Mean	49.840	0.355	49.149	50.557	19754.524	0.000
##						
## Level 1:						
## frlunch.1	0.020	0.229	-0.429	0.467	0.008	0.928
## Residual Var.	21.790	0.999	19.957	23.848	---	---
##						
## Level 2:						
## eslpct	0.022	0.014	-0.006	0.050	2.297	0.130
## frlunch.1	-1.663	0.655	-2.979	-0.373	6.433	0.011
## Residual Var.	1.571	0.705	0.692	3.384	---	---
##						
##						

5) Fit a final model that adds the school-level intervention code. In Blimp, you can automatically create a dummy code by listing condition on the nominal line. To facilitate interpretation, do not center condition, and keep all other predictors centered as they were in previous steps. Using simplified notation from class, the model is as follows. Upload a screen shot of your regression summary table into the BruinLearn quiz. (1 point)

$$psolvepost_{ij} = \beta_0 + \beta_1(psolvepre_{ij}^w) + \beta_2(frlunch_{ij}^w) + \beta_3(psolvepre_j^b) + \beta_4(frlunch_j^b) + \beta_5(eslpct_j) + \beta_6(condition_j) + u_{0j} + \epsilon_{ij}$$

```
p5_mod <- rblimp(
  data = df,
  cluster = 'school',
  center = '
  groupmean = psolvepre frlunch;
  grandmean = psolvepre.mean frlunch.mean',
  nominal = 'frlunch condition',
  model = '
  psolvepost ~ intercept psolvepre psolvepre.mean frlunch frlunch.mean eslpct condition| intercept',
```

```

seed = 90291,
burn = 10000,
iter = 10000
)

```

```

output(p5_mod)

```

```

##
## -----
##
##                               Blimp
##                               4.0.2
##
##       Blimp was developed with funding from Institute of
##       Education Sciences awards R305D150056 and R305D190002.
##
##       Craig K. Enders, P.I. Email: cenders@psych.ucla.edu
##       Brian T. Keller, Co-P.I. Email: btkeller@missouri.edu
##       Han Du, Co-P.I. Email: hdu@psych.ucla.edu
##       Roy Levy, Co-P.I. Email: roy.levy@asu.edu
##
##       Programming and Blimp Studio by Brian T. Keller
##
##       There is no expressed license given.
## -----
##
##
## ALGORITHMIC OPTIONS SPECIFIED:
##
##   Imputation method:           Fully Bayesian model-based
##   MCMC algorithm:              Full conditional Metropolis sampler with
##                               Auto-Derived Conditional Distributions
##
##   Default Priors:
##     Multilevel covariance matrices: Zero matrix, df = -(p + 1) (PRIOR2)
##     Outcome covariance matrices:   Zero matrix, df = -(p + 1) (PRIOR2)
##     Predictor model variances:     Unit sum of squares, df = 2 (XPRIOR1)
##   Chain Starting Values:        Random starting values
##
##
## NOTE: The default prior for regression coefficients
##       in categorical models is 'normal( 0.0, 5.0)'
##
## NOTE: For complete list of priors include 'pinfo' in the options.
##
## NOTE: The following nominal variables have automatically
##       been assigned a reference group.
##
##       'condition' with reference value '0'
##       'frlunch' with reference value '0'
##
##       To change reference group use the following command:
##       NOMINAL: variable_name(reference_value);

```

```

##
##
## BURN-IN POTENTIAL SCALE REDUCTION (PSR) OUTPUT:
##
## NOTE: Split chain PSR is being used. This splits each chain's
##       iterations to create twice as many chains.
##
## Comparing iterations across 2 chains      Highest PSR      Parameter #
##           251 to 500                      1.249              3
##           501 to 1000                     1.064              5
##           751 to 1500                     1.045              3
##          1001 to 2000                     1.065             38
##          1251 to 2500                     1.064             38
##          1501 to 3000                     1.037             28
##          1751 to 3500                     1.098             38
##          2001 to 4000                     1.106             38
##          2251 to 4500                     1.059             38
##          2501 to 5000                     1.076             38
##          2751 to 5500                     1.055             38
##          3001 to 6000                     1.043             38
##          3251 to 6500                     1.015             38
##          3501 to 7000                     1.012             13
##          3751 to 7500                     1.008             38
##          4001 to 8000                     1.006             15
##          4251 to 8500                     1.010              5
##          4501 to 9000                     1.011              5
##          4751 to 9500                     1.013             13
##          5001 to 10000                    1.012              3
##
##
## METROPOLIS-HASTINGS ACCEPTANCE RATES:
##
## Chain 1:
##
## Variable                Type      Probability      Target Value
## frlunch                  latent imp    0.502          0.500
## frlunch                  parameter    0.527          0.500
## psolvepre                latent imp    0.524          0.500
## psolvepre                parameter    0.518          0.500
##
## NOTE: Suppressing printing of 1 chains.
##       Use keyword 'tuneinfo' in options to override.
##
##
## DATA INFORMATION:
##
## Level-2 identifier:      school
## Sample Size:             982
## Level-2 Clusters:       29
## Nominal Dummy Codes:
##
##           condition = condition.1
##           frlunch = frlunch.1
## Missing Data Info:

```

```

##
##           Pattern #:           1
##           miss % -----
##           psolvepost = 0.0      -
##           condition = 0.0      -
##           eslpct = 0.0         -
##           frlunch = 0.0        -
##           psolvepre = 0.0      -
##           -----
##           % 100.0
##
##
## MODEL INFORMATION:
##
##   NUMBER OF PARAMETERS
##     Outcome Models:           9
##     Predictor Models:         22
##
##   PREDICTORS
##     Complete continuous:      eslpct psolvepre
##     Complete nominal:         condition frlunch
##
##   CENTERED PREDICTORS
##     Group Mean Centered:      frlunch psolvepre
##
##   LATENT CLUSTER MEANS
##     Grand Mean Centered:      frlunch psolvepre
##
##   MODELS
##     [1] psolvepost ~ Intercept psolvepre psolvepre.mean[school] frlunch.1
##           frlunch.1.mean[school] eslpct condition.1
##
##
## WARNING MESSAGES:
##
##   No warning messages.
##
##
## MODEL FIT:
##
##   INFORMATION CRITERIA
##
##     Marginal Likelihood
##       DIC2                     5800.814
##       WAIC                     5802.907
##
##     Conditional Likelihood
##       DIC2                     5751.178
##       WAIC                     5763.949
##
##
## IMPUTED DATA SUMMARIES:
##

```

NOTE: Imputed data summaries are averaged across 10000 data sets.

##

##

## Variables	Mean	StdDev	25%	50%	75%	Min
##	-----					
## psolvepost	54.009	5.749	51.000	54.000	58.000	37.000
## condition.1	0.552	0.506	0.000	1.000	1.000	0.000
## eslpct	54.172	22.739	39.000	58.000	67.750	10.000
## frlunch.1	0.801	0.399	1.000	1.000	1.000	0.000
## psolvepre	49.910	4.945	47.000	50.000	53.000	37.000

##

##

##

##

##

CORRELATIONS AMONG RESIDUALS:

##

No residual correlations.

##

##

##

OUTCOME MODEL ESTIMATES:

##

Summaries based on 10000 iterations using 2 chains.

NOTE: Estimate column based on posterior median.

##

##

Outcome Variable: psolvepost

##

Grand Mean Centered: frlunch.1.mean[school] psolvepre.mean[school]

Group Mean Centered: frlunch.1 psolvepre

##

##

## Parameters	Estimate	StdDev	2.5%	97.5%	ChiSq	PValue
##	-----					

Variances:

## L2 : Var(Intercept)	4.033	1.705	1.996	8.616	---	---
------------------------	-------	-------	-------	-------	-----	-----

## Residual Var.	20.202	0.929	18.445	22.144	---	---
------------------	--------	-------	--------	--------	-----	-----

##

Coefficients:

## Intercept	51.473	1.575	48.445	54.645	1068.481	0.000
--------------	--------	-------	--------	--------	----------	-------

## psolvepre	0.461	0.031	0.399	0.523	216.402	0.000
--------------	-------	-------	-------	-------	---------	-------

## psolvepre.mean[school]	0.902	0.388	0.157	1.707	5.475	0.019
---------------------------	-------	-------	-------	-------	-------	-------

## frlunch.1	-0.593	0.394	-1.360	0.183	2.257	0.133
--------------	--------	-------	--------	-------	-------	-------

## frlunch.1.mean[school]	-3.691	3.665	-10.850	3.568	1.022	0.312
---------------------------	--------	-------	---------	-------	-------	-------

## eslpct	0.024	0.023	-0.021	0.069	1.062	0.303
-----------	-------	-------	--------	-------	-------	-------

## condition.1	1.814	0.917	-0.060	3.595	3.889	0.049
----------------	-------	-------	--------	-------	-------	-------

##

Standard Deviations:

## L2 : SD(Intercept)	2.008	0.387	1.413	2.935	---	---
-----------------------	-------	-------	-------	-------	-----	-----

## Residual SD	4.495	0.103	4.295	4.706	---	---
----------------	-------	-------	-------	-------	-----	-----

##

Standardized Coefficients:

```

##      psolvepre                0.367      0.026      0.314      0.418      194.761      0.000
##      psolvepre.mean[school]    0.251      0.099      0.043      0.437       6.165      0.013
##      frlunch.1               -0.038      0.025     -0.087      0.012       2.252      0.133
##      frlunch.1.mean[school]   -0.093      0.089     -0.260      0.086       1.066      0.302
##      eslpct                   0.085      0.080     -0.073      0.238       1.084      0.298
##      condition.1              0.151      0.074     -0.005      0.290       4.033      0.045
##
## Proportion Variance Explained
##   by Coefficients                0.289      0.043      0.206      0.375        ---        ---
##   by Level-2 Random Intercepts    0.118      0.042      0.061      0.222        ---        ---
##   by Level-1 Residual Variation    0.589      0.045      0.491      0.667        ---        ---
##
## -----
##
##
##
## PREDICTOR MODEL ESTIMATES:
##
## Summaries based on 10000 iterations using 2 chains.
## NOTE: Estimate column based on posterior median.
##
##
## Complete predictor: condition
##
## Parameters                Estimate      StdDev      2.5%      97.5%      ChiSq      PValue
## -----
## Grand Mean: condition.1      0.151      0.317     -0.456      0.783      0.246      0.620
##
## Level 2:
##   condition.1 ~ eslpct      -0.023      0.014     -0.052      0.002      2.980      0.084
##   condition.1 ~ frlunch.1   -0.400      0.871     -2.283      1.195      0.251      0.617
##   condition.1 ~ psolvepre    0.132      0.259     -0.383      0.651      0.265      0.606
##   Var(condition.1)          1.000      0.000      1.000      1.000        ---        ---
##
## -----
##
##
## Complete predictor: eslpct
##
## Parameters                Estimate      StdDev      2.5%      97.5%      ChiSq      PValue
## -----
## Grand Mean                54.307      4.966     44.319     64.089     119.379      0.000
##
## Level 2:
##   condition.1              -7.948      4.273    -16.017      0.928      3.361      0.067
##   frlunch.1                 6.632     14.015    -21.462     33.169      0.212      0.645
##   psolvepre                  6.049      3.979     -1.761     13.892      2.354      0.125
##   Residual Var.             365.102    134.053    189.084    712.652        ---        ---
##

```

```

## -----
##
##
## Complete predictor: frlunch
##
## Parameters      Estimate      StdDev      2.5%      97.5%      ChiSq      PValue
## -----
## Grand Mean: frlunch.1      0.993      0.136      0.738      1.272      53.457      0.000
##
## Level 1:
##   frlunch.1 ~ psolvepre      0.001      0.010      -0.019      0.022      0.009      0.923
##   Var(frlunch.1)      1.000      0.000      1.000      1.000      ---      ---
##
## Level 2:
##   frlunch.1 ~ condition.1      -0.067      0.134      -0.313      0.208      0.229      0.632
##   frlunch.1 ~ eslpct      0.003      0.007      -0.011      0.016      0.183      0.669
##   frlunch.1 ~ psolvepre      -0.184      0.088      -0.362      -0.013      4.426      0.035
##   Var(frlunch.1)      0.209      0.090      0.101      0.445      ---      ---
##
## -----
##
##
## Complete predictor: psolvepre
##
## Parameters      Estimate      StdDev      2.5%      97.5%      ChiSq      PValue
## -----
## Grand Mean      49.850      0.408      49.024      50.623      14943.384      0.000
##
## Level 1:
##   frlunch.1      0.023      0.225      -0.416      0.469      0.011      0.917
##   Residual Var.      21.800      1.003      19.965      23.899      ---      ---
##
## Level 2:
##   condition.1      0.183      0.360      -0.556      0.838      0.219      0.640
##   eslpct      0.025      0.016      -0.008      0.056      2.266      0.132
##   frlunch.1      -1.591      0.719      -3.061      -0.201      5.020      0.025
##   Residual Var.      1.537      0.693      0.673      3.315      ---      ---
##
## -----

```

6) Fill in the Rights and Sterba (2019) effect sizes from the four models you fit in the table below, and

upload a screen shot of the table into the BruinLearn quiz. Do the observed changes in the Rights and Sterba effect sizes make theoretical sense? Why or why not? Your answer should clearly explain which source of variation is targeted—or is theoretically eligible to decrease—at each step of the modeling sequence. (2 points)

Effect sizes: * M1 = 0 * M2 = 0.142 * M3 = 0.267 * M4 = 0.289

The observed changes in the Rights and Sterba effect sizes make theoretical sense. In the unconditional model, the effect size is 0 because no predictors are included, so the outcome variance is not explained by any fixed effects. The second model increases to 0.142, reflecting the proportion of within-school variation explained by the level-1 group-mean centered predictors. The third model introduces school means of the level-1 predictors as well as a school-level covariate and we observe an effect size of 0.267. Including these variables explain the between-school variation in the outcome, reflecting a reduction in random intercept variance attributed to school differences. The final model adds the school-level intervention indicator and we observe a modest increase in effect size to 0.289, indicating further reduction in between-school variation aside from what was already explained by the other level-2 covariates.

7) Evaluate the following statement: “The condition coefficient represents the effect of the treatment, controlling for (holding constant / partialing out) pretest problem-solving scores, free/reduced lunch status, and the percentage of ESL students in a school.” Explain precisely which aspect of each variable serves as the control, and why. (2 points)

8) The motivating research question is whether the intervention was effective, controlling for covariates measured at the student and school levels. Using the results from Model 4, report the (a) model-predicted means for the treatment and control groups (both values are obtainable from the fitted model), (b) whether the intervention effect was statistically significant, and (c) an R^2 effect size that represents the unique influence of the intervention variable net the covariates. Provide a brief summary of your conclusion about the effectiveness of the intervention. (2 points)

9) The analysis examples we have used thus far set the number of burn-in iterations to 10,000 and the number of post-burn-in iterations to 10,000. Are these values sufficient for this analysis? In your answer, identify which diagnostic information addresses the adequacy of each value, and report the specific evidence from your output that supports your conclusion. (2 points)

10) To practice with different notational schemes, write the level-1 and level-2 equations for the final model in Equation 4 (i.e., the two-level notation from Raudenbush and Bryk’s book). Perform substitution to obtain the so-called combined-model regression equation. Upload a screen shot of your equations into the BruinLearn quiz. (2 points)